

Abdul Samad

AI/ML Engineer | Software Engineering Student

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PROFESSIONAL SUMMARY

Software Engineering undergraduate (CGPA 3.96/4.00 · 1st Position Holder) with 3 AI/ML internships and a co-authored research paper on automated radiology report generation. Experienced building end-to-end ML pipelines, multi-agent systems, and deep learning models using PyTorch, scikit-learn, FastAPI, and HuggingFace Transformers. Seeking an AI/ML or Software Engineering internship in Islamabad to deploy research-backed solutions to production systems.

EXPERIENCE & INTERNSHIPS

Artificial Intelligence Intern · DecodeLabs | Remote

May 2026 – Present

- Built Python-based automation agents integrating n8n workflow orchestration to handle structured data ingestion and multi-step agentic processing pipelines
- Automated internal data collection and preprocessing routines using Python scripts, reducing manual pipeline setup time and improving data consistency across workflows

Machine Learning Intern · Arch Technologies | Remote

April 2026

- Developed a California Housing Price Prediction model using Random Forest Regressor on 20,640 records, achieving $R^2 = 0.82$ and RMSE of \$48,941 on held-out test data
- Engineered end-to-end preprocessing pipeline with median imputation, standard scaling, and one-hot encoding, enabling clean training with zero manual intervention
- Built an Email Spam Classifier using TF-IDF vectorization (5,000 features) and Multinomial Naive Bayes on an 80/20 split, achieving high precision on unseen messages
- Trained an MLP neural network on MNIST achieving 97%+ digit recognition accuracy across all 10 classes

Machine Learning & Deep Learning Intern · Elevvo | Remote

December 2025

- Implemented data manipulation and model training workflows using NumPy, Pandas, and TensorFlow across 5 structured modules, producing reproducible Jupyter notebooks for each task
- Applied Logistic Regression and K-Means Clustering to real-world datasets, analyzing convergence behaviour and comparing performance across multiple random seeds
- Analyzed student performance datasets using K-Means clustering to identify performance cohorts, presenting findings with structured visualizations in a written report

RESEARCH & PUBLICATIONS

Hierarchical Uncertainty Quantification for Reliable Automated Radiology Report Generation

Abdul Samad et al. · Institute of Management Sciences, Peshawar · Submitted to Journal of Computer Science & Technology (JCST)

- Co-authored research paper proposing a hierarchical uncertainty quantification framework for automated chest X-ray report generation using Vision Transformer (ViT) encoders and GPT-2 decoders
- Designed image feature extraction interfaces linking ViT embeddings to GPT-2 text-generation layers; evaluated model performance using BLEU score and uncertainty calibration metrics
- Validated framework against baseline models on a publicly available radiology dataset; presented results in structured LaTeX with full loss curves and ablation tables

TECHNICAL PROJECTS

AgriGuard 2.0 — Multi-Agent Agricultural Diagnostic System

Python · FastAPI · React · n8n · Multi-Agent Orchestration | github.com/abdulsamad-codes/agriguard

- Architected a multi-agent agricultural diagnostic system with a FastAPI backend and React frontend, migrating from a Streamlit prototype to a production-grade architecture supporting real-time inference
- Designed and implemented multi-agent orchestration workers to handle parallel diagnostic tasks, coordinating custom dataset pipelines and model serving endpoints
- Automated dataset ingestion and preprocessing using Python, enabling seamless data flow between agent workers and the prediction backend with no manual intervention

TECHNICAL SKILLS

Languages: Python · C++ · JavaScript (ES6+) · SQL

ML / AI / Data: scikit-learn · PyTorch · TensorFlow · HuggingFace Transformers · Pandas · NumPy · Matplotlib · Seaborn · Random Forest · Naive Bayes · MLP · K-Means · XGBoost

LLM / Agents: LangChain · LangGraph · n8n Workflow Automation · Agentic AI Patterns · RAG (in progress)

Web & APIs: FastAPI · REST APIs · Streamlit · HTML5 · CSS3

Tools: Git · GitHub · Jupyter Notebook · VS Code · Docker (basic) · Command Line

Foundations: Data Structures & Algorithms · OOP · Feature Engineering · Graph Theory · Probability & Statistics

EDUCATION

Bachelor of Science in Software Engineering

Institute of Management Sciences (IMSciences), Peshawar | 2024 – Present (Expected May 2028) | CGPA: 3.96 / 4.00 | **1st Position Holder**

Relevant Coursework: Data Structures & Algorithms · OOP · Database Systems · Probability & Statistics · Discrete Structures · Software Design · Web Technologies

CERTIFICATIONS

- **Artificial Intelligence Essentials** — University of Pennsylvania (Coursera) | June 2026
- **Supervised Learning with scikit-learn** — DataCamp | April 2026
- **Machine Learning Specialization** — DeepLearning.AI / Coursera (Andrew Ng) | In Progress
- **Deep Learning Specialization** — DeepLearning.AI / Coursera (Andrew Ng) | In Progress
- **Kaggle Learn: Machine Learning & Pandas** — Kaggle | February 2026
- **AI For Everyone** — DeepLearning.AI / Coursera | January 2025

ADDITIONAL INFORMATION

- **Languages:** Urdu (Native) · English (Professional Working Proficiency)
- **Currently Expanding:** XGBoost · Agentic AI · RAG pipelines · LangGraph · Docker
- **Availability:** Immediately available for on-site summer internships in Islamabad or remote positions